Day 25 — Window Operations : Scenario: Detect a sudden increase in transactions during a 10-minute window.

COMMANDS :

```
cd ~/day23-dstreams

cat > src/main/scala/Day25Windows.scala << 'EOF'
import org.apache.spark.SparkConf
import org.apache.spark.streaming.{Seconds, StreamingContext}
import scala.util.Try

object Day25Windows {

  // Spike rule: window count must reach MIN_TXNS and be at least SPIKE_FACTOR x the
previous window
  val MIN_TXNS     = 10
  val SPIKE_FACTOR = 2.0
  private var prevCount = -1L

  def main(args: Array[String]): Unit = {
    val batchSec  = 5
    val windowSec = if (args.length > 0) args(0).toInt else 30   // use 600 for 10 minutes
    val slideSec  = if (args.length > 1) args(1).toInt else 10   // use 30 for the 10-minute run

    // Window and slide must be multiples of the batch interval
    require(windowSec % batchSec == 0 && slideSec % batchSec == 0,
      s"window (windowSec)andslide(slideSec) must be multiples of the batch interval
($batchSec)")

    val conf = new SparkConf().setAppName("Day25Windows").setMaster("local[2]")
    val ssc  = new StreamingContext(conf, Seconds(batchSec))
    ssc.sparkContext.setLogLevel("WARN")
    ssc.checkpoint("/tmp/day25-checkpoint")   // countByWindow needs it

    // Input: ACCOUNT,AMOUNT   e.g. ACC101,250.00
    val lines = ssc.socketTextStream("localhost", 9999)

    val txns = lines.map(_.trim).filter(_.nonEmpty)
      .map(_.split(",")).filter(_.length == 2)
      .flatMap(a => Try(a(1).trim.toDouble).toOption.map(v => (a(0).trim.toUpperCase, v)).toList)

    // 1. countByWindow: number of transactions in the last window, updated every slide
    val windowCount = txns.countByWindow(Seconds(windowSec), Seconds(slideSec))
```

```
    // 2. reduceByKeyAndWindow: rolling sales total per account over the window
    val rollingTotals = txns.reduceByKeyAndWindow(
      (a: Double, b: Double) => a + b, Seconds(windowSec), Seconds(slideSec))

    rollingTotals.foreachRDD { (rdd, time) =>
      println(s"\n--- ROLLING SALES TOTALS, last windowSecs(time) ---")
      val rows = rdd.collect().sortBy(-_._2)
      if (rows.isEmpty) println("(no transactions in window)")
      rows.foreach { case (acc, total) => println(f"$acc%-8s $total%10.2f") }
    }

    // 3. Spike detection: compare this window's count with the previous window's
    windowCount.foreachRDD { (rdd, time) =>
      rdd.collect().foreach { count =>
        println(s"=== WINDOW COUNT: $count transactions in last ${windowSec}s (previous:
$prevCount) ===")
        if (prevCount >= 0 && count >= MIN_TXNS && count >= SPIKE_FACTOR *
math.max(prevCount, 1L)) {
          println(s"!!! ALERT: sudden increase in transactions ($prevCount -> $count) !!!")
        }
        prevCount = count
      }
    }

    ssc.start()
    println(s"Day 25 started: window=windowSecs, slide ={slideSec}s, batch=${batchSec}s. Port
9999 (ACCOUNT,AMOUNT)...")
    ssc.awaitTermination()
  }
}
EOF

cd ~/day23-dstreams
rm -rf /tmp/day25-checkpoint
sbt "runMain Day25Windows"
```

OUTPUTS :

```
[info] --- ROLLING SALES TOTALS, last 30s (1791782415000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782425000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782435000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782445000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782455000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782465000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782475000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782485000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782495000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782505000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782515000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782525000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782535000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782545000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782555000 ms) ---
[info] (no transactions in window)
[info] --- ROLLING SALES TOTALS, last 30s (1791782565000 ms) ---
[info] (no transactions in window)
```